This assignment will not be graded and is only for practice.

Level 1:

1) Let $A = \begin{bmatrix} 0 & 1 & 3 \\ 1 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}$ be a matrix (where the first row denotes the top most row, and the ordering of the rows is in the order top to bottom). Among the given set of options, identify the correct statements.

1 point

- [ ] The first non-zero element in the first row is 3.
- [ ] The first non-zero element in the second row is 1.
- [ ] There is a non-zero element in the third row.
- [ ] Since there is a row with all elements as zero, $det(A) = 0$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The first non-zero element in the second row is 1.

Since there is a row with all elements as zero, $det(A) = 0$.

Let $I_{3\times3}$ denote the identity matrix of order 3. Answer questions 2 and 3 about the set S defined as :

$$S = \left\{ A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, B = \begin{bmatrix} -1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, C = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, D = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \right.$$

$$E = \begin{bmatrix} 0 & 1 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}, F = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, G = \begin{bmatrix} 0 & 1 & 3 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, H = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

$$\left. I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}, J = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, K = I_{3\times3}, L = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \end{bmatrix} \right\}.$$

2) If S_1 is the subset of S consisting of all the matrices in S that are in row echelon form, then choose the correct option from the following. ***1 point***

- [ ] $S_1 = S$
- [ ] $S_1 = \{A, B, C, D, F, G, I\}$
- [ ] $S_1 = \{B, C, D, E, F, G, H, I, J, K, L\}$
- [ ] $S_1 = \{C, D, E, F, G, H, I, J, K\}$
- [ ] $S_1 = \{C, D, E, F, G, H, J, K, L\}$
- [ ] $S_1 = \{C, D, F, G, H, J, K, L\}$
- [ ] Cardinality of S_1 is 7.
- [ ] Cardinality of S_1 is 8.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$S_1 = \{C, D, F, G, H, J, K, L\}$
Cardinality of S_1 is 8.

3) If S_2 is the subset of S consisting of all the matrices in S that are in reduced row echelon form, then choose the correct option from the following. ***1 point***

- [ ] $S_2 = \{A, C, D, F, G, L\}$
- [ ] $S_2 = \{B, C, D, E, F, G, H, J, K\}$
- [ ] $S_2 = \{C, D, G, H, J, K, L\}$
- [ ] $S_2 = \{C, D, H, J, K, L\}$
- [ ] $S_2 = \{C, D, H, I, J, K, L\}$
- [ ] Cardinality of S_2 is 7.
- [ ]

Cardinality of S_2 is 6.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$S_2 = \{C, D, H, J, K, L\}$
Cardinality of S_2 is 6.

4) Consider the following system of linear equations: ***1 point***

$$
\begin{aligned}
0x_1 + x_2 + 0x_3 + 0x_4 &= 1 \\
0x_1 + 0x_2 + 0x_3 + 0x_4 &= 1 \\
x_1 + x_2 + 0x_3 + 0x_4 &= 1 \\
0x_1 + 0x_2 + x_3 + x_4 &= 1.
\end{aligned}
$$

Choose the set of correct options.

- [ ] The system of linear equations has a solution
- [ ] The system of linear equations has no solution.
- [ ] $det(A) = 0$, where A is the coefficient matrix of the given system of linear equations.
- [ ] None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The system of linear equations has no solution.
$det(A) = 0$, where A is the coefficient matrix of the given system of linear equations.

5) Consider a system of linear equations: ***1 point***

$$
\begin{aligned}
0x_1 + x_2 + 0x_3 + 0x_4 &= 1 \\
0x_1 + 0x_2 + x_3 + 0x_4 &= 1
\end{aligned}
$$

Choose the set of correct options.

[Hint: Recall the definitions of independent and dependent variable with respect to reduced row echelon form.]

- [ ] x_1 and x_2 are dependent variables.
- [ ] x_2 is a dependent variable.
- [ ] x_3 and x_4 are independent variables.

☐ x_4 is an independent variable.

No, the answer is incorrect.

Score: 0

Accepted Answers:

x_2 is a dependent variable.
x_4 is an independent variable.

Level 2:

6) Suppose a system of linear equations consists of only one equation and four variables as follows:

$$x_1 + x_2 + x_3 + x_4 = a$$

where a is a constant. Find out the number of independent variables.

[Hint:"Think about" how many variables can be expressed in terms of the other variables in the given system of linear equations?]

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 3

1 point

7) Let $[A|b]$ denote the augmented matrix of the system of linear equations ***1 point***

$$2x_1 + x_2 = 3$$

$$x_1 + 3x_2 = 4$$

where, $A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$, and $b = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$. Let the matrix

$$\left[\begin{array}{cc|c} 1 & 0 & a \\ 0 & 1 & b \end{array}\right]$$

denote the reduced row echelon form of the augmented matrix corresponding to the system of linear equations above. Which of the following option(s) is (are) correct?

- [ ] The values of a and b cannot be determined from the given information.
- [ ] $a = b$ but their exact values cannot be determined from the given information.
- [ ] $a = b = 1$

- [ ] $a = 2$ and $b = 3$
- [ ] The solutions for x_1 and x_2 are not unique.
- [ ] $x_1 = x_2 = 1$, and the system of linear equations has a unique solution.
- [ ] $x_1 = x_2 = 1$ is the solution. However, it is not possible to determine whether the system of equations has a unique solution or not from the given information.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$a = b = 1$

$x_1 = x_2 = 1$, and the system of linear equations has a unique solution.

8) Let $Ax = b$ be a matrix representation of a system of linear equations, where A is a 4×4 matrix, $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$, and $b = \begin{bmatrix} b_1 \\ b_2 \\ 0 \\ 0 \end{bmatrix}$. ***1 point***

Let A be $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$.

Which of the following options are correct?

[Hint: Recall the definitions of independent and dependent variable with respect to reduced row echelon form.]

- [ ] x_1 is dependent on x_2.
- [ ] x_3 is dependent on x_4.
- [ ] x_2 is an independent variable.
- [ ] The solution of the system of linear equations (if it exists) is unique.

- [ ] There exist infinitely many solutions for the given system of linear equations.

No, the answer is incorrect.

Score: 0

Accepted Answers:

x_3 is dependent on x_4.
x_2 is an independent variable.
There exist infinitely many solutions for the given system of linear equations.

Let $A = \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$ be a matrix (where the first row denotes the top most row, and the ordering of the rows is from top to bottom). Consider the system of linear equations given by $Ax = b$ where $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$, and $b = \begin{bmatrix} 4 \\ 3 \\ 0 \end{bmatrix}$. Answer the following questions.

9) Find the number of independent variables.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

10) If $x_1 = 0$ is given, then find out the number of solutions of the given system of linear equations.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

Your score is: 0/10